

Saket Deshmukh

GenAI Engineer — AI Engineer — LLM — RAG — Agentic AI

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SUMMARY

GenAI Engineer with **2.5+ years** building production-grade AI applications using **RAG, agentic workflows, LLM evaluation, and fine-tuning**. Experienced in developing **grounded and guardrailed AI systems** with hybrid retrieval, vector search, tool calling, and self-hosted inference, with end-to-end ownership from **architecture to Kubernetes deployment**.

SKILLS

GenAI: RAG, Agentic AI, LLM Applications, Prompt Engineering, Guardrails, Grounding

LLMs & LLMOps: LLM Evaluation, LLMOps, LangSmith, TruLens, OpenAI API

AI Frameworks: LangChain, LangGraph, FastAPI, REST APIs

Retrieval & Search: FAISS, BM25, pgvector, Hybrid Search, Embeddings, RRF, Semantic Search, Vector Search

Model Serving: vLLM, Self-Hosted Inference, Hugging Face

Programming: Python, JavaScript (Basics), Bash

Databases: PostgreSQL, MySQL

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Helm, Terraform, Ansible

CI/CD & DevOps: GitHub Actions, Jenkins, GitLab CI, Git, Linux

EXPERIENCE

KWAD Corporation Pvt. Ltd. — AI Engineer

Apr 2025 – Present — Pune, India

AI Compliance & Financial Operations Copilot

- Designed and built an **agentic RAG system** using **LangChain** and **LangGraph** to investigate B2B transaction and compliance data across **12 tenants**, orchestrating multi-step workflows with conditional routing, tool calling, retrieval, response generation, and human-in-the-loop escalation.
- Engineered the **RAG pipeline** using chunking, embeddings, and **pgvector**, implementing **grounded responses with source citations**, reducing unsupported/hallucinated answers by an estimated **38%** based on evaluation sampling.
- Designed a **distillation-based fine-tuning pipeline** using **OpenAI's fine-tuning API** to improve structured JSON output reliability, achieving **99% schema compliance** through automated validation.
- Deployed and monitored the system using **LangSmith, Docker, Kubernetes, Helm, and GitHub Actions**; tracing and RAG evaluation metrics supported **9 iterations** of prompt/retrieval improvements with **45% faster mean-time-to-detect** regressions.

Briowin Private Limited — Software Engineer

Feb 2024 – Mar 2025 — Mumbai, India

CatalogSage – AI-Powered Product Catalog Assistant

- Built an **AI-powered RAG product assistant** for natural-language search and recommendations across a **300+ SKU catalog** using Python, FastAPI, MCP, and OpenAI, reducing query resolution time by an estimated **52%** versus manual search.
- Designed a **LangGraph workflow** with conditional routing for input validation, hybrid retrieval, response generation, output validation, and graceful fallback, improving end-to-end request success rate to **94%**.
- Engineered **hybrid retrieval** using **OpenAI embeddings**, **FAISS** semantic search, and **BM25** keyword retrieval, merging results using **Reciprocal Rank Fusion (RRF)**, improving top-k retrieval accuracy by **27%** over a single-method baseline.
- Built a **RAG evaluation pipeline** using **TruLens** to measure groundedness, context relevance, and answer relevance; containerized the service with **Docker** for deterministic CI testing, catching regressions in **18%** of pre-release builds before ship.

PROJECTS

Self-Hosted LLM Inference Platform

AWS EKS, Kubernetes, Terraform, Docker, vLLM

- Built and deployed a **self-hosted LLM inference platform** on **AWS EKS**, using **Docker** and **vLLM** for model serving, with **Terraform** provisioning VPC, IAM, and Kubernetes infrastructure.

AI-Powered SQL Query Optimizer

Python, FastAPI, LangGraph, OpenAI API, SQLGlott, PostgreSQL, Docker

- Built an AI-driven SQL optimization pipeline using **LangGraph**, **OpenAI API**, and **SQLGlott** to analyze PostgreSQL query plans and generate indexing/rewrite recommendations.
- Designed a **safety-validation and result-equivalence layer** to verify query correctness before benchmarking, with measurable execution-time improvements on a **500K+ row** dataset.

EDUCATION

B.E. in Computer Engineering

2019 – 2023 — Pune, India

Savitribai Phule Pune University